

INMO(27th) – 2012

Time – 4 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let ABCD be a quadrilateral inscribed in a circle. Suppose $AB = \sqrt{2 + \sqrt{2}}$ and AB subtend 135° at the centre of the circle. Find the maximum possible area of ABCD.
 2. Let $p_1 < p_2 < p_3 < p_4$ and $q_1 < q_2 < q_3 < q_4$ be two sets of prime numbers such that $p_4 - p_1 = 8$ and $q_4 - q_1 = 8$. Suppose $p_1 > 5$ and $q_1 > 5$. Prove that 30 divides $p_1 - q_1$.
 3. Define a sequence $f_0(x) = 1, f_1(x) = x, (f_n(x))^2 - 1 = f_{n+1}(x)f_{n-1}(x)$, for $n \geq 1$. Prove that each $f_n(x)$ is a polynomial with integer coefficients.
 4. Let ABC be a triangle. An interior point P of ABC is said to be good if we can find exactly 27 rays emanating from P intersecting the sides of the triangle ABC such that the triangle is divided by these rays into 27 smaller triangles of equal area. Determine the number of good points for a given triangle.
 5. Let ABC be an acute angled triangle, let D, E, F be points on BC, CA, AB respectively such that AD is the median, BE is the internal angle bisector and CF is the altitude. Suppose $\angle FDE = \angle C$, $\angle DEF = \angle A$ and $\angle EFD = \angle B$. Prove that ABC is equilateral.
 6. Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be a function satisfying $f(0) \neq 0, f(1) = 0$ and for all $x, y \in \mathbb{Z}$
 $f(xy) + f(x)f(y) = f(x) + f(y)$ and $(f(x - y) - f(0))f(x)f(y) = 0$.
 Then
 (a) Find set of all possible values of the function f .
 (b) If $f(10) \neq 0, f(2) = 0$, find the sets of all integers n such that $f(n) \neq 0$.

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